

Annex 8 — Sector: Plastic Sorting Rejects (France)

Purpose: Sector-specific feedstock profile for plastic sorting rejects — characterisation of non-recyclable plastic packaging waste available in Île-de-France that could supply the Urban Rig’s sorting facility as feedstock, with analysis of economic dynamics (who pays whom) and polymer composition.

Date: March 2026 **Status:** DRAFT — Initial research compilation

OBJECTIVE

To answer the question: **Which identifiable plastic waste streams could we target as dedicated feedstock for the sorting operation, and do their producers pay gate fees to dispose of them or sell them as commodities?**

This annex complements the mixed-waste analyses in Annexes 5 (OMR), 6 (CDG Airport), and 7 (DAE Industrial) by focusing on streams that are predominantly or entirely plastic. These streams are relevant for two reasons: (a) they represent the highest-yield feedstock for pyrolysis oil production (65–80% oil yield vs 23% for mixed waste), and (b) they clarify whether the Urban Rig’s sorting facility can operate as a standalone profit centre by accepting plastic waste at a gate fee.

Every parameter is classified as: **FACT** (measured or from official sources), **HYPOTHESIS** (literature-based estimate), or **ASSUMPTION** (engineering judgment).

PART 1 — THE FRENCH PLASTIC WASTE LANDSCAPE

This part answers: how much plastic waste does France generate, where does it come from, and what happens to it?

1. NATIONAL PLASTIC WASTE GENERATION

1.1 Total Volumes

Parameter	Value	Source
Total plastic waste generated in France	~3.4 million T/year (2020)	ADEME, via Senate Report r20-217; PlasticsEurope 2018 estimate: 3.7 MT
Plastic packaging waste	2.35 million T/year (2020)	Senate Report r22-850 — packaging rose from 2.0 to 2.4 MT between 2010 and 2020
Non-packaging plastic waste	~1.05–1.35 million T/year	Derived: total minus packaging
Plastic recycling rate (France, 2022)	29%	EEA France country profile, 2022
Plastic to incineration (2022)	43%	EEA France country profile, 2022

Parameter	Value	Source
Plastic to landfill (2022)	28%	EEA France country profile, 2022
Plastic NOT recycled	~2.4 million T/year	71% of 3.4 MT = 2.4 MT to incineration + landfill

[FACT] **FACT** — ADEME national statistics, cross-validated by PlasticsEurope and EEA. Data quality: HIGH.

Key insight: Over 2.4 million tonnes of plastic waste per year in France is NOT mechanically recycled. This is the addressable market for alternative value recovery (valorisation — value recovery) routes including pyrolysis.

1.2 Breakdown by Sector

Sector	% of plastic waste	Est. tonnage	Key polymers	Source
Packaging	64%	~2,180 kT	PE, PP, PET, PS	ADEME gisements recyclage chimique
Automotive (VHU)	5%	~170 kT	PP (35%), PU (19%), PA (11%), PVC (9%)	EuRIC position paper; ScienceDirect ELV study
Construction and De-molition	4.5%	~153 kT	PVC (pipes, frames), EPS/XPS (insulation), PE	Frontiers BTP France study
Agriculture	3.6%	~122 kT	LDPE (films), HDPE (pipes), PP (twine)	Adivalor/ADEME
WEEE/DEEE (electronics)	3.5%	~119 kT	ABS, PS, PP, PC-ABS	PMC 8775953; 2.6 MT/yr Europe-wide, France ~4.6% share
Textiles	~2%	~68 kT	PET (polyester), PA (nylon)	EEA textiles data
Furniture	~1%	~34 kT	PP, PE, PU (foam)	ADEME REP furniture
Other / unallocated	~16.4%	~558 kT	Mixed	ADEME residual

[HYPOTHESIS] **HYPOTHESIS** — Sector percentages from ADEME gisement study (2019 data, published 2022). Tonnages derived from 3.4 MT total. Data quality: MEDIUM-HIGH.

2. WHAT HAPPENS TO NON-RECYCLED PLASTIC

2.1 The Two Fates

Of the 2.4 million tonnes of non-recycled plastic per year in France:

Destination	Volume	Cost to producer	Source
Incineration (energy recovery)	~1.46 MT (43%)	€100-150/T gate fee	Zero Waste France incineration — avg €116/T in 2020
Landfill	~0.95 MT (28%)	€80-120/T gate fee + TGAP (General Tax on Polluting Activities) tax	ORDIF TGAP data

[FACT] **FACT** — Incineration cost from Zero Waste France analysis of 2020 data: 1.7 billion EUR for 14.5 million tonnes incinerated = €117/T average. TGAP (General Tax on Polluting Activities / General Tax on Polluting Activities / Taxe Générale sur les Activités Polluantes) increasing to €15-25/T by 2025 on top of gate fees. Data quality: HIGH.

2.2 Rising Cost Pressure

Several regulatory forces are driving the cost of non-recycled plastic disposal upward:

Mechanism	Impact	Timeline	Source
TGAP (General Tax on Polluting Activities / Taxe Générale sur les Activités Polluantes) landfill tax escalation	Landfill gate fee +€15-25/T by 2025	2019-2025	ORDIF TGAP Chronicle
EU plastic packaging tax	€800/T on non-recycled plastic packaging	Since 2021	WTS plastic taxation update 2024
CITEO eco-contribution increase	+15-31% increase on plastic packaging in 2024	2024	Natural Development CITEO 2024
Anti-Waste Law for a Circular Economy (Loi AGECE — Law on Anti-Waste and Circular Economy) targets	100% recycled packaging by 2025 (ambition), end of single-use plastic	2020-2040	AGEC 2025 impact
REP escape rate	40% of REP-subject waste still escapes collection (6.6 MT/yr)	2023	Senate Report on REP (Extended Producer Responsibility / Responsabilité Étendue du Producteur)

[FACT] **FACT** — Regulatory sources from official French government and EU publications. Data quality: HIGH.

Key insight: The economic environment is increasingly hostile to non-recycled plastic disposal. Gate fees are rising, taxes are escalating, and producers face mounting regulatory pressure. This creates a growing market for any alternative that can offer a lower-cost disposal route while qualifying as “recycling” or “value recovery” under EU definitions — exactly what pyrolysis offers.

PART 2 — TARGETABLE PLASTIC-ONLY STREAMS

This part answers: which specific plastic waste streams are available as concentrated feedstock, and what are their economics?

3. STREAM CLASSIFICATION — THE ECONOMIC DIVIDE

All plastic waste streams fall into one of two economic categories, depending on their quality and homogeneity:

3.1 The Two Economic Worlds

Category	Description	Who pays whom	Typical price	Available for UR?
A: Commodity streams	Clean, mono-material, post-industrial. Already has buyers.	Recycler PAYS producer €200-500/T	Positive value	NO — competitive, no gate fee
B: Negative-value streams	Mixed, contaminated, multi-material, or non-recyclable. No buyer.	Producer PAYS for disposal €80-150/T	Negative value	YES — gate fee revenue

[ASSUMPTION] **ASSUMPTION** — Classification based on market dynamics observation. The boundary between A and B is not fixed — it shifts with virgin polymer prices, recycling capacity, and regulation. In 2022 (high prices), more streams were Category A; in 2023 (price collapse), more shifted to Category B.

3.2 The Critical Implication

The Urban Rig’s sorting/pyrolysis operation should target **Category B streams exclusively**. Category A streams are more profitable if sold directly to mechanical recyclers — there is no economic rationale for pyrolysing clean LDPE film that fetches €350/T when it could be sold as-is. The UR’s value proposition is accepting waste that *nobody else wants* and extracting value through thermal conversion.

4. CATEGORY A — COMMODITY STREAMS (NOT PRIMARY TARGETS)

These are documented here for completeness, as they establish the price floor that determines when a stream tips from Category A to Category B.

4.1 Post-Industrial Stretch Film (LDPE)

Parameter	Value	Source
Source sectors	Logistics, warehouses, distribution centres	Plasticsrecyclers.eu flexible films report
European collection volume	5.7 million T/year flexible films collected (2020)	Circular Plastics NL flexible films 2023
Polymer	~95% LDPE/LLDPE	Industry standard
Market price (baled, clean)	€200–400/T (2024)	EEA scrap price data — PE scrap avg €330/T in 2023
IdF warehouse area	17 million m ²	Institut Paris Région (cited in Annex 7)
Est. film generation (IdF)	~30,000 T/year	ASSUMPTION: 1.8 kg/m ² /year × 17M m ² = ~30 kT

[HYPOTHESIS] **HYPOTHESIS** — Collection volume from industry report; IdF estimate derived from warehouse density. MEDIUM confidence.

Verdict: Category A. Already has established buyers (Paprec, Veolia, Suez). Not a target for the UR unless prices collapse below €100/T (making mechanical recycling uneconomic).

4.2 Post-Industrial Production Scrap (PP, HDPE, ABS)

Parameter	Value	Source
Source sectors	Manufacturing, injection moulding, extrusion	General industry knowledge
Market price (clean, colour-sorted)	€300–500/T	Plasticker Central Europe prices
Polymer purity	>98% single polymer	Manufacturers sort at source

Verdict: Category A. Highest-value plastic scrap. Sold directly to compounders/recyclers. Never a UR target.

4.3 Agricultural Film (LDPE)

Parameter	Value	Source
Collection system	Adivalor (EPR scheme)	Marine Litter Solutions — Project RAFU
Volume collected (France)	~35,000 T/year	Adivalor 2011 data; target 100% by 2023

Parameter	Value	Source
Recycling	Dedicated plant (Plasticlean/Calvet, 10,000+ T/yr)	FreshPlaza Adivalor
IdF relevance	LOW — agricultural activity minimal in IdF	Geographic mismatch

Verdict: Category A, handled by dedicated EPR. Not relevant for IdF-based Urban Rig.

5. CATEGORY B — NEGATIVE-VALUE STREAMS (PRIMARY TARGETS)

These are the streams where producers currently **pay** to dispose of their plastic waste, making them ideal feedstock for the Urban Rig at a gate fee.

5.1 Understanding the French Plastic Sorting Chain

Before detailing the individual streams, it is essential to understand how plastic packaging flows through the French recycling system, because two physically distinct “reject” (sorting rejects / refus de tri) streams emerge at different points in the sorting line — one with >95% plastic content, the other mixed material at only 35–45% plastic.

How the two reject streams diverge inside the sorting centre:

A modern sorting centre (centre de tri) processes incoming yellow-bin material through a sequence of mechanical and optical stages. First, ballistic separators divide the flow into 2D (flat/film) and 3D (rigid/volumetric) fractions, while magnets and eddy-current separators extract metals. Then NIR (near-infrared) optical sorters — typically TOMRA AUTOSORT or Pेलenc Mistral units — spectrally scan each item and fire compressed-air jets to eject identified polymers into category-specific bales (TOMRA — sorting technology). These machines achieve >95% purity on their positive-sort output (Antonopoulos et al. 2021 — >99% for PS rigid, ~95% for PE film).

This process produces two types of non-recycled output:

- **Positively sorted plastic with no recycling buyer (→ B1a):** Items the NIR sorter correctly identified and ejected as a specific polymer (PS pot, PE film, opaque PET tray, PP tub, multi-layer pouch), but for which no mechanical recycler currently accepts the material. These items sit in polymer-category bales at >95% plastic purity — the machine confirmed what they are. They are sent to incineration not because they couldn’t be sorted, but because no recycling route exists for them. (refus de tri positifs — sorting rejects that are positively identified)
- **Mixed unidentified residual (→ B1b):** Items the optical sorter could not identify — too small, too contaminated, composite, or obscured. These fall through to the residual conveyor along with soiled paper, textile fragments, organic contamination, and fines. Plastic content is only 35–45% because non-plastic contaminants dominate this stream. (refus de tri mixte — mixed sorting rejects)

The Urban Rig targets the B1a stream specifically — acquiring the positive-sort output that the sorting centre has already confirmed as plastic at >95% purity via NIR spectroscopy.

The sorting chain since the Extension of Sorting Guidelines (ECT — Extension des Consignes de Tri / Extension of Sorting Guidelines, fully deployed 2023):

Stage	What happens	Output	Source
1. Yellow bin	Consumers deposit ALL plastic packaging + paper/cardboard/metal	Mixed recyclable stream	Syndicat Tri Action — ECT
2. Sorting Centre (centre de tri — sorting centre, ~200 in France)	Ballistic separation, magnetic separation, optical sorting, manual sorting	Sorted bales by material type; REJECTS (18-24% of input) = mixed material	COTREP — Plastics sorting in France; La Gazette des Communes
3a. Mechanical recycling	PET bottles, HDPE bottles, some PP → pellets	Recycled resin (~29% of plastic input)	CITEO Data 2024 — plastic recycling rate ~29-30%
3b. Over-sorting Centre (centre de surtri — over-sorting centre, 4 facilities, 100 kT/yr capacity)	Fine optical sorting of “difficult” plastics: PS pots, PET trays, PE films, opaque PET	~90-95% to new recycling streams; 5-10% REJECTS = nearly 100% plastic	CITEO — Over-sorting Centres
3c. Incineration / SRF	All plastic that cannot be mechanically recycled and exceeds over-sorting capacity	Destroyed or converted to solid recovered fuel (SRF — CSR / Combustible de Secours de Récupération, French)	Zero Waste France

[FACT] **FACT** — ECT deployment confirmed by CITEO. Sorting centre count from COTREP. Over-sorting centres (centres de surtri) details from CITEO (4 facilities: Ruffey-lès-Beaune 30 kT/yr, Épinal 25 kT/yr, Mende 15 kT/yr, Thiverval-Grignon 30 kT/yr = 100 kT/yr total, €70M CITEO investment). Data quality: HIGH.

The critical volume gap:

Parameter	Value	Source
Household plastic packaging generated (France)	~1,085 kT/year	COTREP — bottles 435 kT, pots/trays 370 kT, films 280 kT
Plastic packaging recycled	~315 kT/year (~29%)	CITEO 2024
Plastic packaging collected but NOT recycled	~770 kT/year	Derived: $1,085 \text{ kT} \times (1 - 0.29) = \sim 770 \text{ kT}$
Over-sorting centres capacity	100 kT/year	CITEO — 4 centres
Plastic entering system with NO recycling outlet	~670 kT/year	Derived: 770 kT – 100 kT over-sorting capacity = ~670 kT to incineration/SRF

[HYPOTHESIS] **HYPOTHESIS** — Volume gap derived from COTREP generation data and CITEO recycling rate. The 670 kT figure is conservative — it does not include non-packaging plastic waste (additional ~1 MT). MEDIUM-HIGH confidence.

Key insight: Approximately **670,000 tonnes per year** of plastic packaging enters the French sorting system, is positively identified as plastic by NIR optical sorters at >95% purity, but has no mechanical recycling outlet and is sent to incineration or SRF production. This is not material that was merely “collected as plastic” — it was spectrally confirmed as a specific polymer by industrial sorting equipment. It is this NIR-verified stream (B1a), not the general mixed sorting rejects (refus de tri mixte — mixed sorting rejects) (B1b), that represents the highest-quality feedstock for pyrolysis.

5.2 Stream B1a: Plastic Sorting Rejects — Non-Recyclable Plastic Packaging (★★★★★)

This is the single most important feedstock stream for the Urban Rig.

Parameter	Value	Source
Definition	Plastic packaging that is correctly collected and sorted but cannot be mechanically recycled — PS pots/trays, multi-layer films, opaque/coloured PET, mixed PP, contaminated PE films	COTREP recyclability guide
National volume	~670 kT/year (household packaging only)	Derived above — collected plastic minus recycled minus surtri capacity
Est. IdF volume (20% share)	~135,000 T/year	ASSUMPTION: IdF ~20% of national population/consumption
Plastic content	>95%	This material has been positively identified as plastic by NIR optical sorters (TOMRA AUTOSORT, Pellenc Mistral, or equivalent) operating at >95% purity on positive sorts (TOMRA — sorting technology; Antonopoulos et al. 2021 — >99% purity for PS rigid, ~95% for PE film). The material was not merely “collected as plastic” — it was machine-identified, spectrally confirmed as a specific polymer, and ejected into a polymer-category bale. It simply has no mechanical recycling buyer.

Parameter	Value	Source
Polymer mix	PS 25%, PE films 25%, PP 20%, PET (opaque/coloured) 15%, multi-layer 15%	ASSUMPTION based on COTREP data on non-recyclable packaging types
Current disposal	Incineration (€100–150/T) or SRF (Solid Recovered Fuel / CSR) preparation (€40–80/T)	Zero Waste France; FNADE SRF study
Aggregation point	Already aggregated at ~200 sorting centres (centres de tri) across France	COTREP
Gate fee to incinerator	€100–150/T (IdF: €120–150/T due to capacity constraints)	Zero Waste France; ORDIF
SRF production from sorting rejects	~300 kT/year nationally, but 80% of sorting rejects still NOT valorised (converted to value) as SRF	L'Usine Nouvelle — SRF

[HYPOTHESIS] **HYPOTHESIS** — National volume derived from COTREP/CITEO data. IdF share is an estimate. Polymer mix based on COTREP analysis of non-recyclable packaging types. Plastic content (>95%) is based on NIR optical sorter manufacturer specifications for positive-sort purity (TOMRA; Antonopoulos et al. 2021 report >99% purity for PS rigid, ~95% for PE films, ~80% for mixed film fractions). MEDIUM-HIGH confidence.

Why this is the absolute #1 target:

1. **Confirmed >95% plastic content via NIR optical sorting.** Modern sorting centres (centres de tri) use NIR (near-infrared) optical sorters — typically TOMRA AUTOSORT or Pellenc Mistral — that spectrally identify each item's polymer type before ejecting it with compressed air. These machines achieve >95% purity on positively sorted output (TOMRA; Antonopoulos et al. 2021). The B1a stream consists of items the machine correctly identified as plastic (PS, PE film, opaque PET, PP, multi-layer) but for which no mechanical recycler exists. Because the Urban Rig acquires the *positive sort output* — not the mixed residual — the >95% plastic content is an equipment-verified specification, not an assumption. This is fundamentally different from the general mixed sorting rejects (B1b, §5.3), which consist of material the optical sorter *could not identify* or that fell through before reaching the NIR stage.
2. **Massive volume — 135,000 T/year in IdF alone.** The UR at 46,200 T/year would consume only 34% of the IdF supply. There is no risk of feedstock scarcity.
3. **Already aggregated.** The material sits in bales at the ~200 sorting centres across France (and ~20–30 in IdF). No dispersed collection is needed — the logistics infrastructure already exists.
4. **Disposal is a costly problem for sorting centres.** They currently pay €100–150/T to incinerate this material. The UR can offer €80–120/T and still generate substantial margin from pyrolysis oil — because the oil yield from near-100% plastic feed is 65–80% (vs 23% for OMR).

5. **The over-sorting bottleneck creates opportunity.** CITEO has built only 100 kT/year of over-sorting capacity to handle ~770 kT/year of non-recyclable plastic. The remaining ~670 kT/year has no destination other than incineration. The UR positions itself as an alternative value recovery route for this massive surplus.
6. **Even SRF is underserving this market.** Although SRF (Solid Recovered Fuel / CSR — Combustible de Secours de Récupération, French) preparation exists as an intermediate valorisation (value recovery) route, only ~300 kT/year of SRF is produced nationally — meaning 80% of sorting centre rejects are still NOT valorised (converted to value) as SRF and go straight to incineration at €100–150/T (L'Usine Nouvelle — SRF). The UR competes not just against incineration but also against SRF — and wins on economics, because pyrolysis oil at €500/T is far more valuable than the thermal energy SRF produces (€30–50/T equivalent).
7. **Regulatory tailwind.** The Anti-Waste Law for a Circular Economy (Loi AGECE — Law on Anti-Waste and Circular Economy) target of “100% recycled plastic by 2025” is unachievable with current infrastructure. If pyrolysis qualifies as chemical recycling, sorting centres routing material to the UR could count it as “recycled” rather than “incinerated” — dramatically improving their reported recycling rates and CITEO compliance.

5.3 Stream B1b: General Sorting Centre Rejects (Mixed Sorting Rejects — Refus de Tri Mixte)

This is the previously described mixed-material reject stream from sorting centres, now distinguished from the plastic-specific B1a stream.

Parameter	Value	Source
Definition	Material rejected by sorting centres (centres de tri) — too contaminated, mixed, or small for ANY recycling stream (not just plastic)	ADEME Optigede
Reject rate (door-to-door collection)	24.4% of input	ADEME / La Gazette des Communes
Reject rate (voluntary drop-off)	18.4% of input	ADEME via La Gazette des Communes
Number of sorting centres (France)	>350 (excl. inert waste)	ADEME census 2021
Disposal cost for sorting centres	€100–150/T	Zero Waste France — incineration avg €116/T
Est. IdF volume	~300,000–500,000 T/year	ASSUMPTION: IdF processes ~2 MT through sorting centres, 20–25% reject

[HYPOTHESIS] **HYPOTHESIS** — Reject rates from ADEME monitoring. IdF volume is engineering estimate. MEDIUM confidence.

Composition of mixed sorting rejects (refus de tri mixte — mixed sorting rejects):

Material	% of rejects	Source
Mixed plastics (film, rigid, contaminated)	35–45%	Industry characterisation studies
Soiled paper/cardboard	15–25%	Contamination by food/liquids
Textiles / composite	10–15%	Non-recyclable mixed materials
Organic contamination	10–15%	Food waste, liquids
Fines / dust	5–10%	Sub-sieve material

[ASSUMPTION] **ASSUMPTION** — Composition based on published characterisation data from French sorting centres. Varies significantly by centre and collection area. MEDIUM confidence.

Why this remains a strong target (but secondary to B1a): Volume is very high (300,000+ T/year in IdF), material is already aggregated at sorting centres (centres de tri), and gate fees of €80–120/T generate revenue. However, the plastic content of 35–45% is significantly lower than B1a's 90–100%, resulting in lower oil yields per tonne processed. For the UR, B1b is best suited as a supplementary stream when B1a supply is insufficient, or as feedstock for the sorting facility (where the UR can extract the plastic fraction before pyrolysis).

5.4 Stream B2: Multi-Layer and Composite Packaging

Parameter	Value	Source
Definition	Multi-material packaging not amenable to mechanical recycling: metallised film, multi-layer pouches, blister packs, Tetra Pak rejects	Industry classification
Volume (France)	Est. 200,000–400,000 T/year	ASSUMPTION based on ~15% of 2.35 MT packaging waste being multi-layer/composite
Polymer content	40–70% (PE + PP + PET layers)	Tetra Pak composition data — 75% card, 20% PE, 5% Al
CITEO eco-modulation penalty (paid by brand owner)	Up to €1,000/T on difficult plastics	CITEO eco-modulation 2024
Physical disposal gate fee (paid by waste operator)	€100–150/T at incinerator	Standard French waste management

[HYPOTHESIS] **HYPOTHESIS** — Volume estimated from packaging waste total and non-recyclable fraction. MEDIUM confidence.

Important — two distinct costs borne by two different actors:

This stream involves two separate economic penalties that apply at different points in the value chain, and understanding the distinction is critical to the UR's commercial strategy:

1. **CITEO eco-modulation penalty (€1,000/T)** — This is a regulatory surcharge paid by the **packaging brand owner** (e.g., Tetra Pak, Danone, Nestlé) to the French EPR scheme (CITEO/Adelphe) for placing non-recyclable or hard-to-recycle plastic packaging on the French market. It is not a disposal fee — it is levied per kilogram of packaging put on market, regardless of what happens to the waste downstream. The penalty exists to incentivise brands to switch to recyclable formats. Source: Adelphe tariff guide 2024.
2. **Physical disposal gate fee (€100-150/T)** — This is what the **sorting centre or waste operator** (e.g., Paprec, Veolia) pays to the incinerator to physically destroy the reject material. The sorting centre collects the multi-layer packaging, cannot mechanically recycle it, and must pay to dispose of it. This is the cost the UR's proposed gate fee of €70-100/T undercuts.

Why this dual cost structure creates a powerful commercial opportunity:

If pyrolysis qualifies as “chemical recycling” under the EU Packaging and Packaging Waste Regulation (PPWR), the brand owner's CITEO penalty drops to near zero for material routed to the UR — because the packaging is now “recycled” rather than “non-recyclable.” This creates the possibility of a three-way commercial arrangement:

- The **brand owner** pays the UR a premium (€50-150/T) to certify that their packaging was chemically recycled, eliminating their €1,000/T CITEO penalty — a massive net saving
- The **sorting centre** pays the UR a standard gate fee (€70-100/T) to accept the physical material — lower than the €100-150/T they currently pay incinerators
- The **UR** receives revenue from both parties (€120-250/T combined), plus the pyrolysis oil revenue from processing

This makes multi-layer packaging potentially the highest-margin feedstock per tonne, even though the physical gate fee alone (€70-100/T) appears modest.

5.5 Stream B3: Soiled Food Packaging (Retail & Catering)

Parameter	Value	Source
Source	Supermarkets (Carrefour, Leclerc, Auchan etc.), restaurants, catering, food markets	Industry knowledge; Senate Report on REP (Extended Producer Responsibility / Responsabilité Étendue du Producteur)
Types	Food trays (PET, PP, PS), film wrap (LDPE), cups (PS, PP), cutlery	Common waste categories
Contamination level	HIGH — food residue, liquids, grease	Cannot be mechanically recycled without washing
Volume (IdF est.)	50,000-100,000 T/year	ASSUMPTION: IdF = ~20% of French retail sector
Current disposal route	Mixed with OMR → incineration	Not separately collected in most cases
Polymer mix	PE 30%, PP 25%, PET 25%, PS 15%, Other 5%	ASSUMPTION based on food packaging composition

[ASSUMPTION] **ASSUMPTION** — Volume and composition are engineering estimates. The key characteristic is food contamination which makes mechanical recycling uneconomic. LOW-MEDIUM confidence.

Why this is a target: Large-scale retail (grande distribution) is increasingly obligated under the Anti-Waste Law for a Circular Economy (Loi AGECE — Law on Anti-Waste and Circular Economy) to reduce plastic packaging and ensure recyclability. Food-contaminated packaging that cannot be mechanically recycled represents a compliance headache. The UR offers a solution that may qualify as “recycling” under evolving EU definitions, giving retailers a pathway to meet targets.

5.6 Stream B4: WEEE/DEEE (Waste Electrical and Electronic Equipment) Plastics (Waste Electrical and Electronic Equipment)

Parameter	Value	Source
WEEE plastics generated (Europe) France share (~4.6%)	2.6 million T/year ~119 kT/year	PMC 8775953 Derived from population share
Effective recycling rate	~55% of collected stream	BSEF/Sofies study 2020
Not recycled	~45% of collected = ~27 kT/year (France)	Derived: 55% recycled, 45% to disposal
Polymer composition	ABS 30%, PS 25%, PP 20%, PC-ABS 15%, Other 10%	BSEF/Sofies; MDPI review
BFR contamination	~3.5% of WEEE plastics contain BFRs (brominated flame retardants, declining)	BSEF 2023 report
Gate fee	€80–200/T (standard); €200–400/T (BFR+/hazardous)	Industry estimates — BFR+ streams cost significantly more to dispose

[HYPOTHESIS] **HYPOTHESIS** — European data scaled to France. WEEE composition from peer-reviewed studies. MEDIUM confidence.

UR compatibility: WEEE plastics containing brominated flame retardants (BFRs) are fully compatible with the Urban Rig. The UR’s patented catalytic halogen adsorption system filters both chlorine (from PVC) and bromine (from BFRs), meaning BFR-containing WEEE plastics can be processed without restriction. This is a significant competitive advantage — most pyrolysis operators reject BFR-contaminated feedstock, while the UR can accept it at premium gate fees. Note: if BFR content exceeds the 0.2% POPs regulation threshold, the material is classified as hazardous waste under EU regulation and commands gate fees of €200–400/T, making it a particularly lucrative feedstock for the UR.

5.7 Stream B5: End-of-Life Vehicle Plastics (VHU/ASR)

Parameter	Value	Source
ELV plastics generated (France)	~170 kT/year	ADEME gisement study — 5% of 3.4 MT
Automotive Shredder Residue (ASR)	Plastic-rich fraction after metal recovery	ScienceDirect ELV study

Parameter	Value	Source
Polymer composition	PP 35%, PU 19%, PA 11%, PVC 9%, ABS 8%, PE 6%, PC 5%, Other 7%	EuRIC position paper
Current recycling rate	<20% of ELV plastics recycled	WEF circular economy ELV
Thermoplastic recovery from ASR	13.3% overall	ScienceDirect sensor sorting ASR
EU recycled content target (2030)	25% of plastics in new cars from recycled sources	EC ELV Regulation
Gate fee (ASR to disposal)	€100-180/T	Industry estimates for hazardous waste-adjacent material

[HYPOTHESIS] **HYPOTHESIS** — ELV polymer composition from EuRIC and peer-reviewed studies. Recovery rates from recent sensor-sorting research. MEDIUM confidence.

Why this is a strong future target: The new EU ELV Regulation (proposed 2023) mandates 25% recycled content in new vehicles by 2030. This creates a pull-through demand for recycled automotive plastics. PP (35% of ELV plastics) is particularly valuable because it is the dominant automotive polymer. Currently, only 13.3% of thermoplastics are recovered from ASR — the rest goes to incineration or landfill at gate fees of €100-180/T.

5.8 Stream B6: Construction and Demolition (BTP) Plastics

Parameter	Value	Source
Construction and Demolition (BTP — Building and Demolition) total waste (France)	227.5 million T/year (70% of all French waste)	Frontiers BTP study
Plastic fraction in construction/demolition waste	~0.5-1%	Low % but huge base = significant tonnage
Construction and Demolition plastic waste (est.)	~150 kT/year	ADEME gisement — 4.5% of 3.4 MT total plastic waste
Key polymers	PVC 50% (pipes, frames), EPS/XPS 20% (insulation), PE 15% (films), Other 15%	ASSUMPTION based on construction material usage
Recycling rate (Construction/Demolition — BTP general)	70-72% (but plastics much lower)	Frontiers BTP study
7-stream sorting obligation (since 2021)	Mandatory under the 7-stream sorting obligation (since 2021) for construction/demolition professionals: paper, metal, plastic, wood, glass, mineral, plaster	Frontiers BTP study

[HYPOTHESIS] **HYPOTHESIS** — BTP plastics are poorly documented at the detailed polymer level. PVC dominance is well-established for construction. LOW-MEDIUM confidence.

UR compatibility: The high PVC content (50%) of construction and demolition plastic waste is not a constraint for the Urban Rig. The UR's patented catalytic halogen adsorption system captures hydrogen chloride (HCl) released during PVC pyrolysis (see Annex 5, §10.5), enabling the UR to process PVC-rich streams that other pyrolysis operators must reject. This makes construction and demolition plastic waste a viable feedstock without pre-sorting to remove PVC. The captured HCl is neutralised in the filtration system and does not contaminate the oil output. Construction and demolition plastics therefore represent an underserved niche where the UR has a clear technological advantage.

5.9 Stream B7: Textile-Origin Plastics

Parameter	Value	Source
Textile waste generated (France)	~800,000 T/year	ADEME — collection in 2021: 244,000 T, representing ~30% of arisings
Synthetic fibre content	~35–40% polyester (PET), ~10% nylon (PA), ~5% acrylic	EEA Textiles and the Environment
Plastic-equivalent tonnage	~280–360 kT/year synthetic fibres	Derived: 800 kT × 35–45% synthetic
Recycled fraction	Very low (<1% fibre-to-fibre recycling)	EEA textiles report
Current disposal	Mostly incineration or landfill (after resale/donation share)	Industry standard
REP textile (France, since 2020)	Mandatory EPR for textiles via Refashion (eco-organisme — eco-organization)	Law on Anti-Waste and Circular Economy / Loi AGECL

[HYPOTHESIS] **HYPOTHESIS** — Tonnages from ADEME and EEA European data. Synthetic fraction well-documented. MEDIUM confidence.

Why this is an emerging target: Fast fashion waste is growing rapidly and is overwhelmingly non-recyclable. Polyester (PET-family) and nylon (PA) fibres are chemically identical to their packaging counterparts and produce the same pyrolysis yields (45% oil for PET, 45% for PA — see Annex 5 §10.4 and §10.7). The REP textile scheme (Refashion) is generating collection volumes faster than recycling capacity can absorb them, creating a growing stockpile of negative-value material.

PART 3 — ECONOMICS AND PRICING

This part answers: for each Category B stream, what gate fee can the UR charge, and what is the total revenue potential?

6. GATE FEE BENCHMARKS

6.1 Current Disposal Costs (What Producers Pay Today)

Disposal route	Gate fee (EUR/T)	TGAP (General Tax on Polluting Activities) tax	Total cost to producer	Source
Incineration (standard)	€90–120	€0–15	€90–135	Zero Waste France — avg €116/T
Incineration (IdF premium)	€120–150	€15–25 (2025)	€135–175	ORDIF; IdF higher due to capacity con- straints
Landfill (ISDND)	€60–90	€25–65 (2025 target)	€85–155	ORDIF TGAP
CSR (Combustible de Secours de Récupération) / SRF production	€40–80	€0	€40–80	ADEME CSR (Com- bustible de Secours de Récupéra- tion) data — lower cost al- ternative
Hazardous waste incineration (BFR+)	€200–400	Included	€200–400	Industry standard for POPs- contaminated waste

[FACT] **FACT** — Gate fees from official French sources. TGAP trajectory from published law. Data quality: HIGH.

6.2 Proposed UR Gate Fees (Category B Streams)

The UR can undercut incineration by 15–30% while still generating positive margin:

Stream	Current disposal cost	Proposed UR gate fee	UR discount vs incin.	Rationale
B1a: Plastic sorting rejects	€100-150/T	€80-120/T	-20%	Near-100% plastic, highest oil yield, same logistics as B1b
B1b: General sorting rejects (refus de tri — sorting rejects)	€100-150/T	€80-120/T	-20%	High volume, steady supply, es- tablished logistics
B2: Multi- layer packag- ing	€100-150/T + CITEO penalty	€70-100/T	-30%	Regulatory credit value if classified as recycling
B3: Soiled food packag- ing	€100-150/T	€80-110/T	-20%	Food con- tamination not a problem for pyrolysis
B4: WEEE (Waste Electri- cal and Elec- tronic Equip- ment) plastics (incl. BFR)	€80-400/T	€90-200/T	-15 to -50%	Higher plastic content = higher oil yield; UR halogen filter handles BFRs
B5: ASR / VHU plastics	€100-180/T	€90-140/T	-15%	Growing regulatory pressure (ELV reg.)
B6: BTP plastics (incl. PVC)	€80-120/T	€60-90/T	-25%	UR halogen filter handles PVC; no pre-sorting required

Stream	Current disposal cost	Proposed UR gate fee	UR discount vs incin.	Rationale
B7: Textile synthetics	€80-130/T	€70-100/T	-20%	Growing surplus from REP textile

[ASSUMPTION] **ASSUMPTION** — Proposed gate fees are engineering and commercial estimates based on undercutting current disposal costs. MEDIUM confidence. Actual gate fees would be negotiated contract-by-contract.

7. RECLAIMED MATERIAL PRICES (COMMODITY VALUE)

When the UR sorts and processes Category B streams, the recovered materials have commodity value:

7.1 Current European Market Prices

Material	Price range (EUR/T)	Base case	Source
Mixed plastic scrap (average)	€250-470	€330	EEA circularity metrics — 2023 EU avg EEA — PE scrap 2023 avg €330/T, virgin €1,444/T Plasticker market; 2022 peak ~€400+ Plasticsrecyclers.eu HDPE/PP report Plasticker; sensitive to virgin price Plasticsrecyclers.eu PET report Limited market, EPS performs better Restricted market due to chlorine ScrapMonster EU index LME-linked
PE scrap (mixed quality)	€250-400	€330	
Sorted LDPE film bales	€200-400	€350	
Sorted HDPE	€250-500	€450	
Sorted PP	€200-450	€400	
Sorted PET (clear flake)	€300-700	€600	
Sorted PS	€100-250	€200	
PVC (sorted, non-BTP)	€50-150	€100	
Ferrous scrap (HMS 1&2)	€200-350	€300	
Aluminium scrap	€800-1,500	€1,200	

Material	Price range (EUR/T)	Base case	Source
Glass cullet	€15-40	€25	Low value, high transport cost
Paper/Cardboard (OCC)	€60-120	€100	REVIPAP index France
Pyrolysis oil (ex-waste)	€400-600	€500	ICIS PPO index; depends on crude

[HYPOTHESIS] **HYPOTHESIS** — Prices from EEA, Plastics, and industry indices. Markets are volatile. MEDIUM confidence. 2022 was a historic high; 2023 saw significant correction. 2024 prices have partially recovered.

Key observation on price volatility: In 2022, average plastic scrap reached €467/T (EEA data) — a historic peak driven by the Chinese import ban aftermath, pandemic recovery demand, and regulatory push. In 2023, prices collapsed as virgin polymer oversupply from China and the USA flooded the market. The base case uses 2023-2024 stabilised prices, which are conservative.

PART 4 — VOLUME ASSESSMENT AND UR CAPACITY

This part answers: how much of these streams could the UR realistically capture?

8. ADDRESSABLE MARKET IN ÎLE-DE-FRANCE

8.1 Volume Summary by Stream

Stream	Est. IdF volume (T/yr)	Plastic content	Effective plastic (T/yr)	Confidence
B1a: Plastic sorting rejects	~135,000	90-100%	121,500-135,000	MEDIUM-HIGH
B1b: General sorting rejects (mixed — refus de tri)	300,000-500,000	35-45%	105,000-225,000	MEDIUM
B2: Multi-layer packaging	40,000-80,000	40-70%	16,000-56,000	LOW-MEDIUM

Stream	Est. IdF volume (T/yr)	Plastic content	Effective plastic (T/yr)	Confidence
B3: Soiled food packag- ing	50,000–100,000	60–80%	30,000–80,000	LOW
B4: WEEE plastics	10,000–25,000	80–95%	8,000–24,000	MEDIUM
B5: VHU/ASR plastics	15,000–30,000	50–70%	7,500–21,000	MEDIUM
B6: Con- struc- tion/Demolition (BTP — Building and Demoli- tion) plastics (incl. PVC)	15,000–30,000	70–90%	10,500–27,000	LOW
B7: Textile synthet- ics	20,000–40,000	90–100%	18,000–40,000	LOW- MEDIUM
TOTAL	585,000–940,000		316,500–608,000	

Note on B1a vs B1b overlap: There is partial overlap between B1a and B1b — some of the plastic in B1b rejects is the same material that would otherwise be classified as B1a if it were sorted more finely. However, the distinction matters commercially: B1a can be sourced as already-sorted plastic bales (higher purity, higher value), while B1b requires the UR's own sorting facility to extract the plastic fraction. In practice, the UR would negotiate with sorting centres for both streams — B1a as premium feedstock for direct pyrolysis, B1b as general input for the sorting + pyrolysis chain.

[ASSUMPTION] **ASSUMPTION** — IdF volumes are engineering estimates based on national data scaled by IdF population/economic share (~20%). Actual volumes depend on collection infrastructure and commercial contracts. LOW-MEDIUM confidence overall.

8.2 UR Capacity Context

Parameter	Value
UR annual throughput	46,200 T/year (at 100% capacity)
Total Category B available (IdF)	585,000–940,000 T/year
B1a alone (IdF)	~135,000 T/year
UR share of B1a supply	34%
UR share of total Category B market	5–8%
If 100% B1a feed: plastic throughput	~43,900 T/year of plastic (95% of 46,200 T)

Parameter	Value
If 100% B1a feed: oil production	~28,600 T/year (62% oil yield)

Key finding: The UR could fill its entire capacity from B1a alone, consuming only 34% of IdF supply. With B1a as feedstock, oil production nearly triples compared to OMR (28,600 T vs ~10,600 T), because the feed is 90–100% plastic rather than 15%. Supply is not a constraint — the constraint is plant capacity.

PART 5 — CONCLUSIONS AND RECOMMENDATIONS

9. STREAM PRIORITISATION

Based on the analysis in Parts 2–4, we rank the Category B streams by attractiveness for the Urban Rig:

9.1 Priority Matrix

Rank	Stream	Gate fee	Volume	Plastic content	Logistics	Score
1	B1a: Plastic sorting rejects (non- recyclable plastic packag- ing)	€80–120/T	Very high (135 kT/yr IdF)	90–100%	Excellent (aggregated at sorting centres)	★★★★★+
2	B1b: General sorting rejects (mixed — refus de tri)	€80–120/T	Very high (300– 500 kT/yr IdF)	35–45%	Excellent (aggregated at sorting centres)	★★★★★
3	B2: Multi- layer packag- ing	€70–100/T	High	40–70%	Good (aggregated at packaging producers)	★★★★★
4	B3: Soiled food packag- ing	€80–110/T	High	60–80%	Medium (dispersed at retail sites)	★★★★★

Rank	Stream	Gate fee	Volume	Plastic content	Logistics	Score
5	B5: VHU/ASR plastics	€90-140/T	Medium	50-70%	Good (aggregated at shredder sites)	★★★★
6	B4: WEEE plastics (incl. BFR)	€90-200/T	Low- Medium	80-95%	Good (via DEEE recyclers)	★★★★
7	B7: Textile syn- thetics	€70-100/T	Medium (grow- ing)	90-100%	Medium (via Refashion (EPR scheme for textiles) hubs)	★★★
8	B6: Con- struc- tion/Demolition plastics (incl. PVC)	€60-90/T	Medium	70-90%	Poor (dispersed at demolition sites)	★★★

Note on B1a vs B1b: The distinction is critical. B1a is plastic that was correctly collected and sorted AS plastic but has no mechanical recycling outlet (~670 kT/year nationally, ~135 kT in IdF). It is 90-100% plastic by weight. B1b is the general mixed-material reject from the sorting process (sorting errors / erreurs de tri — sorting errors, contaminated material, fines), which is only 35-45% plastic. Both are available at the same sorting centres, but B1a is vastly superior as pyrolysis feedstock due to its near-pure plastic content.

Note on halogen-containing streams: The PVC risk column has been removed from this matrix. The UR's patented catalytic halogen adsorption system captures both chlorine (from PVC) and bromine (from brominated flame retardants), meaning all streams above are fully compatible with the UR regardless of PVC or BFR content. This is a key competitive differentiator — most pyrolysis operators reject PVC-rich or BFR-contaminated feedstock, limiting their addressable market. The UR can accept these streams at full or premium gate fees.

9.2 Recommended Supply Strategy

Phase 1 (Year 1-3): Non-recyclable plastic packaging (B1a) as core feedstock - Contract with 5-10 sorting centres (centres de tri — sorting centres) in IdF specifically for their non-recyclable plastic bales (B1a) at €90-110/T - This stream alone (~135 kT/year in IdF) can fill UR capacity nearly 3× over — the challenge is not volume but contract negotiation - Supplement with multi-layer packaging (B2) from packaging producers at €80-100/T - Because B1a is 90-100% plastic, the UR's pre-sort requirement is minimal — material can go almost directly to pyrolysis after shredding

Phase 2 (Year 3-5): Diversify into higher-value streams - Add soiled food packaging from large-scale retail (grande distribution — large-scale retail) contracts (Carrefour, Leclerc etc.) - Add VHU/ASR as EU recycled content mandates create supply push - Target BFR-contaminated WEEE plastics at premium gate fees (€200-400/T) — the UR's halogen filtration gives exclusive access to this high-margin niche - Accept construction and demolition (BTP) plastics including PVC-rich demolition waste — no pre-sorting needed thanks to chlorine filtration - Begin textile synthetic acceptance as Refashion collection volumes outpace recycling capacity

Phase 3 (Year 5+): Premium positioning - If pyrolysis qualifies as “chemical recycling” under EU regulation, reposition as a compliance solution - Charge premium gate fees (€120–150/T) to producers needing to demonstrate recycling of non-recyclable plastics - Position as the only pyrolysis operator in IdF accepting halogenated plastics (PVC, BFR) — a market segment with no competition - Develop CITEO/REP eco-contribution credit trading

10. ECONOMIC IMPACT ON THE BUSINESS MODEL

10.1 Gate Fee Revenue from Category B Streams

Using UR throughput of 46,200 T/year at weighted average gate fee of €95/T:

Scenario	Gate fee	Annual gate revenue	Impact vs OMR model (€120/T gate)
Conservative (sorting rejects only)	€85/T	€3,927,000	-29% vs OMR
Base case (mixed Category B)	€95/T	€4,389,000	-21% vs OMR
Optimistic (with compliance premium)	€110/T	€5,082,000	-8% vs OMR

[ASSUMPTION] **ASSUMPTION** — Gate fees are commercial estimates. Actual revenue depends on contract negotiations. MEDIUM confidence.

10.2 Combined Revenue: Gate Fee + Sorted Material Sales + Pyrolysis Oil

The key advantage of Category B streams vs OMR is **higher plastic content**, which means more pyrolysis oil per tonne processed. Comparison across three scenarios:

Revenue component	OMR (Annex 5, 15% plastic)	B1b: General sorting rejects (40% plastic)	B1a: Plastic sorting rejects (95% plastic)
Gate fee per tonne	€120	€95	€95
Oil yield (% of input)	23%	~35%	~62%
Oil revenue per tonne (at €500/T)	€115	€175	€310
By-product revenue per tonne	~€15	~€15	~€10
Total revenue per tonne	€250	€285	€415

[HYPOTHESIS] **HYPOTHESIS** — Oil yield for B1a: 95% plastic × 65% oil yield per plastic = 61.8%, rounded to 62%. For B1b (40% plastic): $0.40 \times 65\% + 0.60 \times 11\% = 32.6\%$, conservative 35%. MEDIUM confidence.

Key finding: The B1a stream (non-recyclable plastic packaging) generates **€415/T total revenue** — 66% more than OMR (€250/T) and 46% more than general sorting rejects (€285/T). This is because at 95% plastic content, the oil yield approaches pure-plastic levels (62% vs 23% for OMR). Even with a lower gate fee (€95 vs €120/T), the pyrolysis oil revenue more than compensates. **B1a is by far the most profitable feedstock per tonne for the Urban Rig.**

At full capacity (46,200 T/year), a B1a-only feed would generate: - Gate fee revenue: $46,200 \times €95 = \mathbf{€4.39M/year}$ - Oil revenue: $46,200 \times 0.62 \times €500 = \mathbf{€14.32M/year}$ - By-product revenue: ~€0.46M/year - **Total: ~€19.17M/year** vs ~€11.55M/year for OMR — a **66% increase**

11. DATA QUALITY SUMMARY

Parameter	Classification	Confidence
French plastic waste total (3.4 MT)	[FACT] FACT	HIGH — ADEME/PlasticsEurope
Sector breakdown (64% packaging, etc.)	[HYPOTHESIS] HYPOTHESIS	MEDIUM-HIGH — ADEME gisement study
Recycling rate (29%)	[FACT] FACT	HIGH — EEA
Incineration cost (€116/T avg)	[FACT] FACT	HIGH — Zero Waste France analysis
TGAP (General Tax on Polluting Activities) tax trajectory	[FACT] FACT	HIGH — published law
Sorting rejects reject rate (18–24%)	[FACT] FACT	HIGH — ADEME monitoring
Sorting rejects composition (mixed)	[ASSUMPTION] ASSUMPTION	MEDIUM — characterisation studies
B1a volume (~670 kT national, ~135 kT IdF)	[HYPOTHESIS] HYPOTHESIS	MEDIUM-HIGH — derived from COTREP/CITEO data
B1a plastic content (90–100%)	[FACT] FACT	HIGH — by definition, sorted as plastic
Centres de surtri capacity (100 kT/yr, 4 sites)	[FACT] FACT	HIGH — CITEO official data
EEA scrap prices (€330/T PE, etc.)	[FACT] FACT	HIGH — Eurostat/EEA data
WEEE polymer composition	[HYPOTHESIS] HYPOTHESIS	MEDIUM — peer-reviewed studies
ELV polymer composition	[HYPOTHESIS] HYPOTHESIS	MEDIUM — EuRIC/academic
IdF volume estimates	[ASSUMPTION] ASSUMPTION	LOW-MEDIUM — scaled from national data
Proposed gate fees	[ASSUMPTION] ASSUMPTION	MEDIUM — based on undercutting existing costs
Oil yield for 40% plastic feed	[HYPOTHESIS] HYPOTHESIS	MEDIUM — interpolated from Annex 5
UR halogen filtration (Cl + Br)	[FACT] FACT	HIGH — patented catalytic adsorption system (UR Recycling Solutions, pp. 14–15)

12. RECOMMENDATIONS FOR THE BUSINESS PLAN

1. **Target non-recyclable plastic packaging (B1a) as the primary feedstock — not general sorting rejects (refus de tri).** France's sorting centres (centres de tri) produce ~670 kT/year of plastic that has been collected and sorted but has no mechanical recycling outlet, of which ~135 kT/year is in IdF. This material is 90–100% plastic, already aggregated at sorting centres, and generates €415/T total revenue (gate fee + oil) vs €250/T for OMR — a 66% improvement. The UR at 46,200 T/year would consume only 34% of IdF supply.
2. **Do not compete for Category A (commodity) plastic streams.** Clean post-industrial film and production scrap are more valuable sold to mechanical recyclers. The UR's competitive advantage is in accepting waste that is too contaminated, mixed, or problematic for mechanical recycling.
3. **Build the commercial strategy around “the disposal problem.”** Every Category B stream exists because someone has a waste they cannot economically recycle and must pay to dispose of. The UR offers a lower-cost disposal route with the added regulatory benefit of qualifying as thermal treatment / chemical recycling.
4. **Monitor the EU classification of pyrolysis as “recycling” closely.** If pyrolysis of waste plastic is officially classified as recycling (rather than “energy recovery”) under the EU Waste Framework Directive and PPWR (Packaging and Packaging Waste Regulation), the commercial value proposition transforms: producers can claim recycling credits, CITEO/REP contributions drop, and the UR can charge premium gate fees for “recycling” certificates alongside the physical processing fee.
5. **Leverage the UR's halogen filtration as a competitive moat.** The UR's patented catalytic halogen adsorption system captures both chlorine (from PVC) and bromine (from brominated flame retardants), enabling the UR to accept PVC-rich streams (construction/demolition at 50% PVC) and BFR-contaminated WEEE plastics that competing pyrolysis operators must reject. This expands the addressable market to include all seven Category B streams without restriction and allows the UR to command premium gate fees on hazardous-adjacent material (€200–400/T for BFR+ WEEE).
6. **B1a transforms the business model — and reduces the need for pre-sorting.** The standalone sorting business model (see UR-Standalone-Sorting-Business-Model.xlsx) assumed OMR-composition input requiring extensive sorting. With B1a at 90–100% plastic content, the UR's sorting facility becomes largely unnecessary for this stream — the material is already plastic. This reduces OPEX (fewer sorting staff, less equipment wear) while dramatically increasing revenue (€415/T vs €250/T). The sorting facility remains valuable for B1b (general sorting rejects / refus de tri) and other mixed streams, but B1a could be routed directly to shredding and pyrolysis.
7. **Prioritise testing B1a (non-recyclable plastic packaging) on the UR.** A single characterisation + pyrolysis trial run with actual plastic sorting rejects would validate the 90–100% plastic content, confirm the polymer mix (PS/PE/PP/PET distribution), and measure the actual oil yield (expected 60–65%). This single data point would anchor the financial model on measured data and, if confirmed, would transform the UR's business case — shifting projected revenue from €11.55M/year (OMR basis) to ~€19M/year (B1a basis).

13. References

French Plastic Waste Data

1. **ADEME** — National waste statistics; waste resources study for chemical/physicochemical recycling (2019 data, published 2022), [librairie.ademe.fr](https://www.ademe.fr). Cited for: total plastic waste 3.4 MT, sector breakdown (§1), REP furniture data, SRF costs, sorting rejects rates, Optigede monitoring.
2. **Senate Report r20-217** — [senat.fr](https://www.senat.fr). Cited for: total French plastic waste volume (§1.1).
3. **PlasticsEurope** (2018) — Plastic waste generation estimate 3.7 MT. Cited for: cross-validation of ADEME total (§1.1).
4. **EEA (European Environment Agency)** — France country profile 2022, eea.europa.eu. Cited for: plastic recycling rate 29%, incineration 43%, landfill 28% (§1.1).
5. **EEA** — Circularity metrics, average plastic scrap prices (2023), eea.europa.eu. Cited for: PE scrap €330/T, mixed plastic scrap €250-470/T (§8).
6. **EEA** — *Textiles and the Environment* report. Cited for: synthetic fibre content in textile waste (§6.4).

Sorting and Recycling Infrastructure

7. **COTREP** — *Plastics sorting and recycling in France*, cotrep.fr; recyclability guide. Cited for: sorting chain stages, household packaging volumes 1,085 kT/year, non-recyclable plastic definition (§4, §5).
8. **CITEO** — Recycling data 2024, citeo.com; over-sorting centres (centres de surtri), citeo.com. Cited for: plastic recycling rate ~29%, 4 over-sorting centres (100 kT/year, €70M investment), eco-modulation penalties (§4, §5, §7).
9. **La Gazette des Communes** — Sorting error rates, lagazettedescommunes.com. Cited for: reject rates 18.4-24.4% (§6.1).
10. **Natural Development** — CITEO eco-contribution 2024 tariffs, naturaldevelopment.fr. Cited for: +15-31% increase on plastic packaging, eco-modulation penalties up to €1,000/T (§2, §7).

Disposal Cost Sources

11. **Zero Waste France** — Incineration cost analysis, zerowaste-france.org. Cited for: average incineration cost €116/T, gate fee ranges €100-150/T (§1.3, §6.1, §8).
12. **FNADE** — SRF (Solid Recovered Fuel — CSR in French) economic model study, fnade.org. Cited for: SRF preparation costs €40-80/T (§8).
13. **ORDIF** — Île-de-France tipping fee data. Cited for: IdF premium on incineration costs (§5.4).

Sector-Specific Sources

14. **EuRIC** — Position paper on recycled content in plastics, euric-aisbl.eu. Cited for: ELV polymer composition (§6.3).
15. **ScienceDirect** — ELV study, [sciencedirect.com/S2352550921002621](https://www.sciencedirect.com/S2352550921002621). Cited for: VHU/ASR polymer data (§6.3).
16. **Adivalor / ADEME** — Agricultural plastic recycling, via freshplaza.com. Cited for: agricultural plastic volumes 122 kT/year (§1.2).
17. **Refashion** — French EPR eco-organisme for textiles. Cited for: textile collection scheme context (§6.4, §9).

Regulatory Sources

18. **Anti-Waste Law for a Circular Economy** (Loi AGECE, Loi n° 2020-105 du 10 février 2020), via globalclimateinitiatives.com. Cited for: 100% recycled plastic target, REP textile mandate (§6.4, §12).
19. **EU PPWR (Packaging and Packaging Waste Regulation)** — Adopted 2024. Cited for: chemical recycling classification implications (§7, §12).

20. **TGAP** — General Tax on Polluting Activities / Taxe Générale sur les Activités Polluantes (increasing trajectory). Cited for: disposal cost escalation (§1.3).

Urban Rig Technical Sources

21. **Urban Rig Recycling Solutions ENG** — OneWorld Corporation product brochure (pp. 14-15). Cited for: patented catalytic halogen adsorption system (Cl + Br) (§9).

End of Annex 8